Retrieval-Augmented Generation (RAG) Architecture Overview
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Retrieval-Augmented Generation combines an external knowledge retriever with an LLM.
By retrieving relevant passages before generation, RAG reduces hallucinations and grounds claims in real
Dense vector search is effective for semantic matching, but struggles on rare tokens and acronyms.

Lexical search using BM25Okapi indexes exact term frequencies and document lengths.
Reciprocal Rank Fusion combines vector search and BM25 search rankings without needing raw score no
The RRF formula is 1 / (k + rank), where k=60 is standard.

Cross-Encoders perform joint attention over the question and passage pair.
This improves ranking precision and MRR at the cost of higher latency.
On a 16 GB RAM laptop, a compact model like cross-encoder/ms-marco-MiniLM-L-6-v2 balances accurac

Citations must be synthesized directly from chunk metadata (source filename and page number).
The system must never rely on LLM hallucinations for page citations.
When documents lack sufficient evidence, the assistant states that the supplied documents do not provide